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Supporting Material

Kinetics and Regulation of Mammalian NADH-ubiquinone Oxidoreductase (Complex I)

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Supplementary Materials for “Detailed Kinetics and Regulation of Mammalian NADH-ubiquinone Oxidoreductase”

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Kinetic Constants Definition by Rate Constants

$$V_1 = \frac{k_1 k_2 k_3 k_4}{k_1 k_3 k_4 + k_1 k_4 k_{-2} + k_2 k_3 k_4 + k_1 k_2 k_4 + k_2 k_3 k_{-4} + k_1 k_2 k_3}$$

$$V_2 = \frac{k_{-1} k_{-2} k_{-3} k_{-4}}{k_{-1} k_{-2} k_{-3} + k_{-2} k_{-3} k_{-4} + k_{-1} k_{-2} k_{-4} + k_2 k_{-3} k_{-4} + k_{-1} k_{-3} k_{-4} + k_{-1} k_{-2} k_4}$$

$$K_{eq} = \frac{k_1 k_2 k_3 k_4}{k_{-1} k_{-2} k_{-3} k_{-4}} \frac{K_p K_Q}{K_H K_A K_B}$$

$$K_{mA} = K_A \frac{k_2 k_3 (k_4 + k_{-4})}{k_1 k_3 k_4 + k_1 k_4 k_{-2} + k_2 k_3 k_4 + k_1 k_2 k_4 + k_2 k_3 k_{-4} + k_1 k_2 k_3}$$

$$K_{mB} = K_B \frac{k_1 k_4 (k_2 + k_{-2})}{k_1 k_3 k_4 + k_1 k_4 k_{-2} + k_2 k_3 k_4 + k_1 k_2 k_4 + k_2 k_3 k_{-4} + k_1 k_2 k_3}$$

$$K_{mH} = K_H \frac{k_2 k_3 k_4 + k_2 k_3 k_{-4}}{k_1 k_3 k_4 + k_1 k_4 k_{-2} + k_2 k_3 k_4 + k_1 k_2 k_4 + k_2 k_3 k_{-4} + k_1 k_2 k_3} = \frac{K_{mA}}{K_A} K_H$$

$$K_{mP} = K_P \frac{k_{-3} k_{-4} (k_2 + k_{-2})}{k_{-1} k_{-2} k_{-3} + k_{-2} k_{-3} k_{-4} + k_{-1} k_{-2} k_{-4} + k_2 k_{-3} k_{-4} + k_{-1} k_{-3} k_{-4} + k_{-1} k_{-2} k_4}$$

$$K_{mQ} = K_Q \frac{k_{-1} k_{-2} (k_4 + k_{-4})}{k_{-1} k_{-2} k_{-3} + k_{-2} k_{-3} k_{-4} + k_{-1} k_{-2} k_{-4} + k_2 k_{-3} k_{-4} + k_{-1} k_{-3} k_{-4} + k_{-1} k_{-2} k_4}$$

$$K_{ia} = K_A \frac{k_{-3} k_{-4}}{k_{-3} k_{-4} + k_1 k_4 + k_1 k_{-3}}$$

$$K_{ib} = K_B \frac{k_{-1}k_{-2}}{k_{-1}k_{-2} + k_{-1}k_3 + k_2k_3}$$

$$K_{ip} = K_P$$

$$K_{iq} = K_Q \frac{k_1k_4}{k_{-3}k_{-4} + k_1k_4 + k_1k_{-3}}$$

$$K_{ih} = K_H$$

In our parameterization, we assumed that K_{mH} equals K_{ih} , we can derive that $K_{mA} = K_A$. Based on the definitions, we also have $K_{ih} = K_H$ and $K_{ip} = K_P$. We assume that rate constant k_3 depends on membrane potential as $k_3 = k_3^0 \exp(-4F\Delta\Psi / RT)$ to account for the free energy cost of pumping 4 protons across the mitochondrial inner membrane; here k_3^0 is the rate constant when membrane potential is zero.

Computational Model of Mitochondrial Oxidative Phosphorylation

This section presents a simplified version of our computational model of mitochondrial respiratory system and oxidative phosphorylation (1) (also see 2, 3-5). The basic components of the original model include reactions at complexes I, III, IV and V, substrate transporters ANT (adennine nucleotide translocase) and PHT (phosphate-hydrogen cotransporter), and cation fluxes across the IMM, including the fluxes via the KHE (K^+/H^+ exchanger) and passive H^+ and K^+ permeations. For clarity, the simplified governing model equations are presented below and the associated model parameter values are presented in Table S1.

Mitochondrial Transport and Reaction Fluxes:

Dehydrogenase $NAD^+ \rightleftharpoons NADH + H^+$

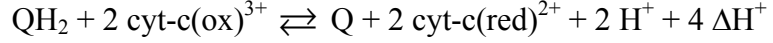
$$J_{DH} = X_{DH} \left(\frac{1 + [Pi]_x / k_{Pi,1}}{1 + [Pi]_x / k_{Pi,2}} \right) (r_{DH} [NAD^+]_x - [NADH]_x)$$

The model do not account for the kinetics of the TCA cycle and other NADH-producing reactions (1-3). Instead, the rate of NADH production from NAD^+ is modeled using a phosphate-dependent phenomenological dehydrogenase flux expression.

Complex I $H^+ + NADH + Q \rightleftharpoons NAD^+ + QH_2 + 4 \Delta H^+$

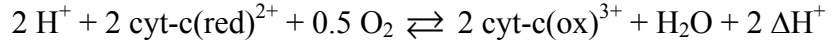
$$J_{CI} = X_{CI} Flux_{CI}$$

where $Flux_{CI}$ is the flux of complex I as shown in Equation (19) in the text. A total of 4 protons are pumped each time this reaction proceeds.

Complex III

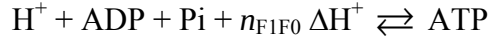
$$J_{\text{C3}} = X_{\text{C3}} \left(\frac{1 + [\text{Pi}]_x / k_{\text{Pi,3}}}{1 + [\text{Pi}]_x / k_{\text{Pi,4}}} \right) \left(\frac{\exp\left(-(\Delta G_{\text{o,C3}} + 4\Delta G_{\text{H}} + 2RT \ln([\text{H}^+]_x / 10^{-7}) - 2F\Delta\Psi)\right) / 2RT}{\times [\text{cyt-c(ox)}^{3+}] [\text{QH}_2]^{1/2} - [\text{cyt-c(red)}^{2+}] [\text{Q}]^{1/2}} \right)$$

The model accounts for a phosphate-dependent control of the complex III reaction. Where $\Delta G_{\text{H}} = F\Delta\Psi + RT \ln([\text{H}^+]_e / [\text{H}^+]_x)$ is the proton motive force, or the change in Gibb's free energy associated with the pumping of protons from the matrix side to the extra-matrix side of the IMM. A total of 4 protons are pumped from the matrix side to the extra-matrix side of the IMM each time this reaction proceeds.

Complex IV

$$J_{\text{C4}} = X_{\text{C4}} \left(\frac{[\text{O}_2]}{k_{\text{O}_2} + [\text{O}_2]} \right) \left(\frac{[\text{cyt-c(red)}^{2+}]}{\text{cyt-c}_{\text{tot}}} \right) \left(\frac{\exp\left(-(\Delta G_{\text{o,C4}} + 2\Delta G_{\text{H}} - 2RT \ln([\text{H}^+]_x / 10^{-7}))\right) / 2RT}{\times [\text{cyt-c(red)}^{2+}] [\text{O}_2]^{1/4} - e^{+F\Delta\Psi/RT} [\text{cyt-c(ox)}^{3+}]} \right)$$

A total of 2 protons are pumped from the matrix side to the extra-matrix side of the IMM each time this reaction proceeds (oxygen consumption).

Complex V (F₁F₀-ATPase)

$$J_{\text{F1F0}} = X_{\text{F1F0}} \left(\frac{\exp\left(-(\Delta G_{\text{o,F1F0}} - n_{\text{F1F0}} \Delta G_{\text{H}} - RT \ln([\text{H}^+]_x / 10^{-7}))\right) / RT}{\times (K_{\text{MgADP}} / K_{\text{MgATP}}) [\text{mADP}]_x [\text{Pi}]_x - (1 \text{ M}) [\text{mATP}]_x} \right)$$

A total of n_{F1F0} (= 3) protons enter the matrix through the proton channel of F₁F₀-ATPase each time this reaction proceeds (ATP synthesis).

Magnesium Binding

$$J_{\text{MgATPx}} = X_{\text{MgA}} ([\text{fATP}]_x [\text{Mg}]_x - K_{\text{MgATP}} [\text{mATP}]_x)$$

$$J_{\text{MgADPx}} = X_{\text{MgA}} ([\text{fADP}]_x [\text{Mg}]_x - K_{\text{MgADP}} [\text{mADP}]_x)$$

$$J_{\text{MgATPi}} = X_{\text{MgA}} ([\text{fATP}]_i [\text{Mg}]_i - K_{\text{MgATP}} [\text{mATP}]_i)$$

$$J_{\text{MgADPi}} = X_{\text{MgA}} ([\text{fADP}]_i [\text{Mg}]_i - K_{\text{MgADP}} [\text{mADP}]_i)$$

[fATP] and [fADP] denote the concentration of ATP and ADP that are not bound to magnesium ion in the matrix and IM space; [mATP] and [mADP] denote the magnesium-bound species.

NADH Binding

We assume simple first-order binding of NADH, with the bound fraction of the binding protein $f_b = X_{\text{bound}} / X_0$ which can be computed as $f_b = [\text{NADH}]_{\text{free}} / (K_n + [\text{NADH}]_{\text{free}})$. The amount of bound NADH is computed $[\text{NADH}]_{\text{bound}} = X_0 f_b$. Here X_0 is protein binding capacity in the matrix, and K_n is the protein-NADH binding dissociation constant. Solving for free NADH as a function of total NADH, we have

$$[\text{NADH}]_{\text{free}} = \frac{1}{2} \left([\text{NADH}]_{\text{total}} - K_n - X_0 + \sqrt{([\text{NADH}]_{\text{total}} - K_n - X_0)^2 + 4K_n[\text{NADH}]_{\text{total}}} \right)$$

Adenine Nucleotide Translocase (ANT)

$$J_{\text{ANT}} = X_{\text{ANT}} \left(\frac{[\text{fADP}]_i}{k_{\text{ADP}} + [\text{fADP}]_i} \right) \left(\frac{[\text{fADP}]_i}{[\text{fADP}]_i + [\text{fATP}]_i e^{-0.35F\Delta\Psi/RT}} - \frac{[\text{fADP}]_x}{[\text{fADP}]_x + [\text{fATP}]_x e^{+0.65F\Delta\Psi/RT}} \right)$$

The exchange of ATP for ADP via the ANT is electrogenic, so the flux is dependent on the IMM potential $\Delta\Psi$. In addition, the exchange is regulated (activated) by the extra-matrix free ADP concentration.

Adenylate Kinase (AK)



$$J_{\text{AKi}} = X_{\text{AK}} (K_{\text{AK}} [\text{ADP}]_i [\text{ADP}]_i - [\text{AMP}]_i [\text{ATP}]_i)$$

Phosphate-Hydrogen Cotransporter (PHT)

$$J_{\text{PHT}} = X_{\text{PHT}} \left(\frac{[\text{H}_2\text{PO}_4^-]_i [\text{H}^+]_i - [\text{H}_2\text{PO}_4^-]_x [\text{H}^+]_x}{[\text{H}_2\text{PO}_4^-]_i + k_{\text{PHT}}} \right)$$

where $[\text{H}_2\text{PO}_4^-]_i = [\text{H}^+]_i [\text{Pi}]_i / ([\text{H}^+]_i + k_{\text{dH}})$, $[\text{H}_2\text{PO}_4^-]_x = [\text{H}^+]_x [\text{Pi}]_x / ([\text{H}^+]_x + k_{\text{dH}})$ and $[\text{H}^+]_i = [\text{H}^+]_e$.

Potassium Hydrogen Exchanger (KHE)

$$J_{\text{KHE}} = X_{\text{KHE}} ([\text{K}^+]_i [\text{H}^+]_x - [\text{K}^+]_x [\text{H}^+]_i); \quad [\text{K}^+]_i = [\text{K}^+]_e \text{ and } [\text{H}^+]_i = [\text{H}^+]_e$$

Passive Proton Permeation (Leak)

$$J_{\text{Hleak}} = X_{\text{Hleak}} \Delta\Psi \left(\frac{[\text{H}^+]_i \exp(+F\Delta\Psi/RT) - [\text{H}^+]_x}{\exp(+F\Delta\Psi/RT) - 1} \right); \quad [\text{H}^+]_i = [\text{H}^+]_e$$

Passive Potassium Permeation (Leak)

$$J_{\text{Kleak}} = X_{\text{Kleak}} \Delta\Psi \left(\frac{[\text{K}^+]_i \exp(+F\Delta\Psi/RT) - [\text{K}^+]_x}{\exp(+F\Delta\Psi/RT) - 1} \right); \quad [\text{K}^+]_i = [\text{K}^+]_e$$

Passive Substrate Transport Fluxes

$$J_{\text{ATPt}} = \gamma p_A ([\text{ATP}]_e - [\text{ATP}]_i)$$

$$J_{\text{ADPt}} = \gamma p_A ([\text{ADP}]_e - [\text{ADP}]_i)$$

$$J_{\text{AMPt}} = \gamma p_A ([\text{AMP}]_e - [\text{AMP}]_i)$$

$$J_{\text{Pit}} = \gamma p_{\text{Pi}} ([\text{Pi}]_{\text{e}} - [\text{Pi}]_{\text{i}})$$

Dynamic Mass Balance Equations:

The following differential equations constitute the simplified model of mitochondrial electron transport system and oxidative phosphorylation.

$$\begin{aligned} d\Delta\Psi/dt &= (+4J_{\text{C1}} + 2J_{\text{C3}} + 4J_{\text{C4}} - n_{\text{F1F0}}J_{\text{F1F0}} - J_{\text{ANT}} - J_{\text{Hleak}} - J_{\text{Kleak}})/C_{\text{IMM}} \\ d[\text{H}^+]_{\text{x}}/dt &= \frac{[\text{H}^+]_{\text{x}}}{\beta_{\text{H}}} (+J_{\text{DH}} - 5J_{\text{C1}} - 2J_{\text{C3}} - 4J_{\text{C4}} + (n_{\text{F1F0}} - 1)J_{\text{F1F0}} + 2J_{\text{PHT}} + J_{\text{Hleak}} - J_{\text{KHE}})/W_{\text{x}} \\ d[\text{K}^+]_{\text{x}}/dt &= (+J_{\text{KHE}} + J_{\text{Kleak}})/W_{\text{x}} \\ d[\text{Mg}^{2+}]_{\text{x}}/dt &= (-J_{\text{MgADP}} - J_{\text{MgATP}})/W_{\text{x}} \\ d[\text{NADH}]_{\text{x}}/dt &= (+J_{\text{DH}} - J_{\text{C1}})/W_{\text{x}} \\ d[\text{QH}_2]_{\text{x}}/dt &= (+J_{\text{C1}} - J_{\text{C3}})/W_{\text{x}} \\ d[\text{cyt-c}(\text{red})^{2+}]_{\text{i}}/dt &= (+2J_{\text{C3}} - 2J_{\text{C4}})/W_{\text{i}} \\ d[\text{ATP}]_{\text{x}}/dt &= (+J_{\text{F1F0}} - J_{\text{ANT}})/W_{\text{x}} \\ d[\text{ADP}]_{\text{x}}/dt &= (-J_{\text{F1F0}} + J_{\text{ANT}})/W_{\text{x}} \\ d[\text{mATP}]_{\text{x}}/dt &= (+J_{\text{MgATPx}})/W_{\text{x}} \\ d[\text{mADP}]_{\text{x}}/dt &= (+J_{\text{MgADPx}})/W_{\text{x}} \\ d[\text{Pi}]_{\text{x}}/dt &= (-J_{\text{F1F0}} + J_{\text{PHT}})/W_{\text{x}} \\ d[\text{ATP}]_{\text{i}}/dt &= (+J_{\text{ATPt}} + J_{\text{ANT}} + J_{\text{AKi}})/W_{\text{i}} \\ d[\text{ADP}]_{\text{i}}/dt &= (+J_{\text{ADPt}} - J_{\text{ANT}} - 2J_{\text{AKi}})/W_{\text{i}} \\ d[\text{AMP}]_{\text{i}}/dt &= (+J_{\text{AMPt}} + J_{\text{AKi}})/W_{\text{i}} \\ d[\text{mATP}]_{\text{i}}/dt &= J_{\text{MgATPi}}/W_{\text{i}} \\ d[\text{mADP}]_{\text{i}}/dt &= J_{\text{MgADPi}}/W_{\text{i}} \\ d[\text{Pi}]_{\text{i}}/dt &= (+J_{\text{Pit}} - J_{\text{PHT}})/W_{\text{i}} \end{aligned}$$

Where W_{x} and W_{i} are the matrix and IM space water volumes, and β_{H} is the proton buffering capacity of the matrix space.

The following state variables are computed from the total concentration pools based on mass conservations:

$$\begin{aligned} [\text{NAD}^+]_{\text{x}} &= \text{NAD}_{\text{tot}} - [\text{NADH}]_{\text{x}} \\ [\text{Q}] &= \text{Q}_{\text{tot}} - [\text{QH}_2] \\ [\text{cyt-c}(\text{ox})^{3+}] &= \text{cyt-c}_{\text{tot}} - [\text{cytC}(\text{red})^{2+}] \\ [\text{ADP}]_{\text{x}} &= \text{A}_{\text{tot}} - [\text{ATP}]_{\text{x}} \end{aligned}$$

Table S1: Parameter values in the simplified model of mitochondrial electron transport system and oxidative phosphorylation (“mito” denotes mitochondria). Most parameter values were estimated based on the experimental data of Bose *et al.* (6) in isolated cardiac mitochondria. For details, see the references (1-5).

Parameter	Description	Values	Units	Reference
<i>Physicochemical/Thermodynamic Parameters:</i>				
RT	Gas constant times temperature	2.5775	kJ mol^{-1}	_a
F	Faraday’s constant	0.096484	$\text{kJ mol}^{-1} \text{mV}^{-1}$	_a
$\Delta G_{0,C1}$	Standard free energy, complex I	-69.37	kJ mol^{-1}	_b
$\Delta G_{0,C3}$	Standard free energy, complex III	-32.53	kJ mol^{-1}	_b
$\Delta G_{0,C4}$	Standard free energy, complex IV	-122.94	kJ mol^{-1}	_b
$\Delta G_{0,F1F0}$	Standard free energy, F_1F_0 -ATPase	36.03	kJ mol^{-1}	_b
<i>Anatomical Parameters:</i>				
W_x	Matrix water fraction	0.6435	$(\text{l water}) (\text{l mito})^{-1}$	_c
W_i	IMS water fraction	0.0715	$(\text{l water}) (\text{l mito})^{-1}$	_c
γ	Outer membrane area per mito volume	5.99	μm^{-1}	_d
<i>Mitochondrial Model Parameters:</i>				
$k_{Pi,1}$	Dehydrogenase/Pi parameter	0.1512	mM	_e
$k_{Pi,2}$	Dehydrogenase/Pi parameter	0.7147	mM	_e
r_{DH}	Dehydrogenase phenomenological parameter	4.4406	Unitless	_e
X_{DH}	Dehydrogenase activity	0.1076	$\text{mol s}^{-1} \text{M}^{-1} (\text{l mito})^{-1}$	_e
X_{C1}	Complex I activity	0.55×10^3	$\text{mol s}^{-1} \text{M}^{-2} (\text{l mito})^{-1}$	_e
X_{C3}	Complex III activity	8.0953	$\text{mol s}^{-1} \text{M}^{-3/2} (\text{l mito})^{-1}$	_e
X_{C4}	Complex IV activity	3.18×10^{-5}	$\text{mol s}^{-1} \text{M}^{-1} (\text{l mito})^{-1}$	_e
X_{F1F0}	F_1F_0 -ATPase activity	80.26	$\text{mol s}^{-1} \text{M}^{-1} (\text{l mito})^{-1}$	_e
X_{ANT}	ANT activity	9.0×10^{-3}	$\text{mol s}^{-1} (\text{l mito})^{-1}$	_e
n_{F1F0}	H^+ stoichiometry for F_1F_0 -ATPase	3.0	unitless	_f
X_{PHT}	Pi^-/H^+ cotransporter activity	1.78×10^5	$\text{mol s}^{-1} \text{M}^{-1} (\text{l mito})^{-1}$	_e
k_{PHT}	Pi^-/H^+ cotransporter parameter	0.3451	mM	_e
X_{KHE}	K^+/H^+ antiporter activity	2.83×10^9	$\text{mol s}^{-1} \text{M}^{-2} (\text{l mito})^{-1}$	_e
X_{Hleak}	Proton leak activity	289.33	$\text{mol s}^{-1} \text{M}^{-1} \text{mV}^{-1} (\text{l mito})^{-1}$	_e
X_{Kleak}	Potassium leak activity	0.0	$\text{mol s}^{-1} \text{M}^{-1} \text{mV}^{-1} (\text{l mito})^{-1}$	_e
$k_{Pi,3}$	Complex III/Pi parameter	0.3128	mM	_e
$k_{Pi,4}$	Complex III/Pi parameter	3.2	mM	_e
p_{Pi}	Outer mitochondrial membrane permeability to inorganic phosphate	327.0	$\mu\text{m sec}^{-1} (\text{l mito})^{-1}$	_g
p_A	Outer mitochondrial membrane permeability to nucleotides	85.0	$\mu\text{m sec}^{-1} (\text{l mito})^{-1}$	_h
k_{ADP}	ANT Michaelis-Menten constant	3.5×10^{-6}	M	_g
k_{O2}	Kinetic constant for complex IV	1.2×10^{-4}	M	_g

β_H	Matrix proton buffering capacity coefficient	0.0043	M	_i
C_{IMM}	Effective capacitance of IMM	6.75×10^{-6}	$\text{mol (l mito)}^{-1} \text{ mV}^{-1}$	_j
K_{AK}	AK equilibrium constant	0.4331	Unitless	_b
<i>Fixed Concentrations Pools:</i>				
NAD_{tot}	Total matrix NAD concentration	2.97	$\text{mmol (l matrix water)}^{-1}$	_g
UQ_{tot}	Total matrix ubiquinol concentration	1.35	$\text{mmol (l matrix water)}^{-1}$	_g
$cyt-c_{tot}$	Total IMS cytochrome-c concentration	2.70	$\text{mmol (l IMS water)}^{-1}$	_g
$[Mg^{2+}]_{tot}$	Total matrix Mg^{2+} concentration	5.0	$\text{mmol (l matrix water)}^{-1}$	_k
<i>Binding Constants:</i>				
K_{MgATP}	Mg-ATP binding constant	24×10^{-6}	M	_g
K_{MgADP}	Mg-ADP binding constant	347×10^{-6}	M	_g
k_{dH}	$H^+ - H_2PO_4^-$ dissociate constant	1.778×10^{-7}	M	_l

_a Standard physicochemical/thermodynamic constants

_b Computed from the thermodynamic data tabulated in Alberty (7)

_c Taken from Vinnakota and Bassingthwaighe (8)

_d Taken from Munoz *et al.* (9)

_e Estimations from Beard (1, 2) and Wu *et al.* (3).

_f Taken from Tomashek & Brusilow (10)

_g Taken from Vendelin *et al.* (11)

_h Taken from Lee *et al.* (12)

_i Taken from Kapus *et al.* (13)

_j Taken from Gentet *et al.* (14)

_k Taken from Bose *et al.* (14)

_l Taken from Alberty (7)

Application to Complex I function in the intact mitochondria

Figure S1 illustrates comparison between the new integrated model and experimentally measured values of $NADH^{2-}$, membrane potential, oxygen consumption, matrix pH and cytochrome C reduced fraction from Bose *et al.* (6).

Shown in Panel A are model simulations of steady-state $NADH^{2-}$ concentration (normalized to NAD_{tot}) as a function of external inorganic phosphate $[Pi]_e$. The two curves correspond to two different values of external $[ADP]_e$ concentration, 0 mM (dashed line) and 1.3 mM (solid line), as indicated in the figure. Data from Bose *et al.* (6), are shown as circles (0 mM $[ADP]_e$) and triangles (1.3 mM $[ADP]_e$). Figure S1 B illustrate the experimental measurements and model simulated the rate of oxygen consumption (MV_{O_2}) values for the same experimental conditions as described for $NADH^{2-}$ curves in Panel A. The nonzero value of MV_{O_2} when substrate concen-

tration goes to zero corresponds to the basal oxygen consumption in mitochondrial respiratory system necessary to maintain the proton motive energy. Figure 7C presents the model simulation and experimental measurement of the membrane potential $\Delta\Psi$ for the same experimental conditions. Panels D and E presents the model simulation and experimental measurements of the cytochrome C reduced fraction and the matrix pH value for the same conditions. The poor fit to the state-3 membrane potential (1.3 mM $[\text{ADP}]_e$) at low phosphate concentration is consistent with previous analyses (1), and is potentially due to an experimental artifact. Since phosphate is a tri-carboxylic acid (TCA) cycle substrate, at zero phosphate concentration both the membrane potential and the NADH should go to zero, as demonstrated in a previous model analysis (4).

Panel F shows the predicted ratio of the mitochondria $[\text{NADH}^{2-}]_{\text{free}}$ to $[\text{NADH}^{2-}]_{\text{total}}$ in the experiments, revealing a ratio between 84% and 90% which is consistent with the result reported by Tischler et al. (15). The assumption $k_3 = k_3^0 \exp(-4F\Delta\Psi / RT)$ is validated by the fact that we are unable to produce reasonable matches between these data and model prediction when the membrane potential dependence is incorporated into other rate constants. These results may be reproduced using a more complex model with all the TCA cycle reactions (4) which was developed based on this biophysical model of the mitochondrial respiratory system and oxidative phosphorylation (1). (These results are not illustrated.)

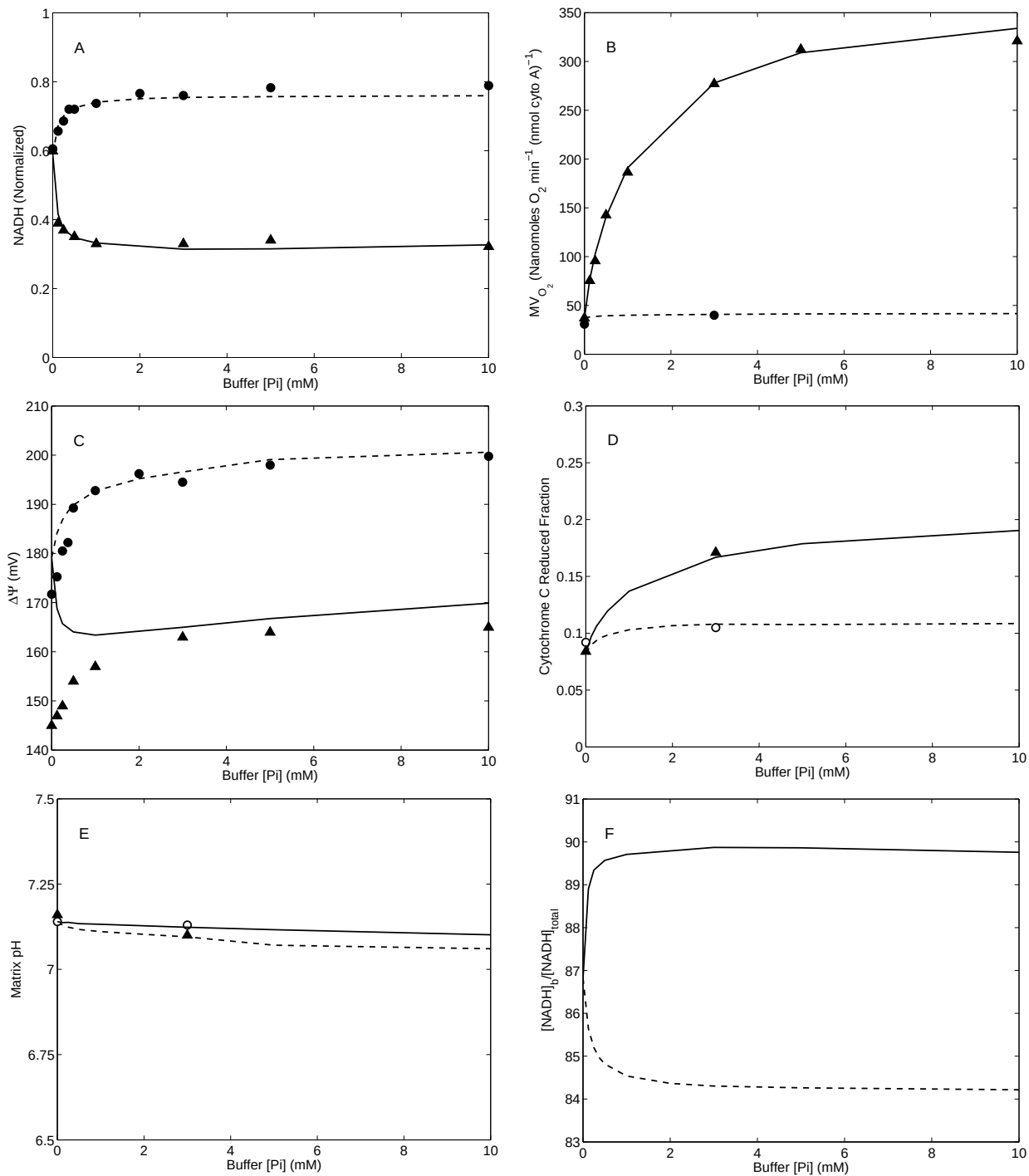


Figure S1: (A) Normalized matrix NADH^{2-} concentration as a function of buffer inorganic phosphate (Pi) concentration is plotted for two experimental cases of resting mitochondria (state 2) and active mitochondria (state 3). (B) Rate of oxygen consumption. (C) Membrane potential $\Delta\Psi$. (D) Cytochrome C reduced level. (F) Ratio of mitochondrial $\text{NADH}^{2-} / \text{NADH}^{2-}_{\text{total}}$ as a function of buffer inorganic phosphate concentration. For all panels, symbols ● and ▲ represent data obtained in the presence of $[\text{ADP}] = 0$ (state 2) and $[\text{ADP}] = 1.3 \text{ mM}$ (state 3) in the buffer. Dashed line and solid line are model predictions for state 2 and state 3, respectively.

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